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Smart Beta Multifactor Construction Methodology: *Mixing versus Integrating*

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Equity factor premiums have been extensively researched and documented. Due to rapid advancements in smart beta product offerings in recent years, exposure to premiums beyond the traditional market beta is now widely available at very low cost through transparent rule-based indexing strategies. Since the early 2000s, smart beta products that focus on a single factor, such as value, low volatility, or quality, have emerged as viable and attractive investment options. Each factor is expected to deliver a premium over the long run, but none is guaranteed to perform well at all times. During and after the global financial crisis, for instance, the U.S. value factor registered a cumulative 12.0% loss in 2007 and 2008, and the U.S. momentum factor recorded a loss of 52.6% in 2009.[1] Seeking to diversify their active risk exposures, many investors have accordingly turned to strategies that exploit multiple factors.

The rising popularity of multifactor smart beta strategies raises practical questions. As Harvey, Liu, and Zhu [2015] pointed out, academics have identified hundreds of factors. How can investors choose the robust and implementable ones (see Beck et al. [2016])? Once they have settled on a promising set of factors, what is the optimal allocation (see Alighanbari and Chia [2016])? Should they maintain a static exposure to the selected factors, or can they further improve returns by means of dynamic reallocations (see Arnott, Beck, and Kalesnik [2016])? The questions surrounding factor selection and allocation have been widely debated.

Our research focuses on a practical issue that has received less attention: the trade-offs market participants should consider when choosing between investment vehicles with different approaches to building multifactor portfolios. Specifically, we study and compare two distinct methodologies. The first one, henceforth called the *integrating* approach, is a process that searches the universe of eligible securities for those that have desirable exposures to all target factors. The second, hereafter the *mixing* approach, is a two-step process of identifying stand-alone single-factor portfolios and allocating assets across them to achieve the desired exposure to multiple factors. Intuitively, the integrating approach has the advantage of selecting securities from the entire opportunity set, whereas the mixing approach has the obvious benefits of flexibility in factor allocation and simplicity in performance attribution.

The few researchers who have compared the two methodologies generally support the integrating approach. (We discuss their studies in the literature review.) Yet we observe that smart beta investors increasingly favor the mixing approach. As an indication of adoption rates, Exhibit 1 shows the total amount of exchange-traded fund (ETF)

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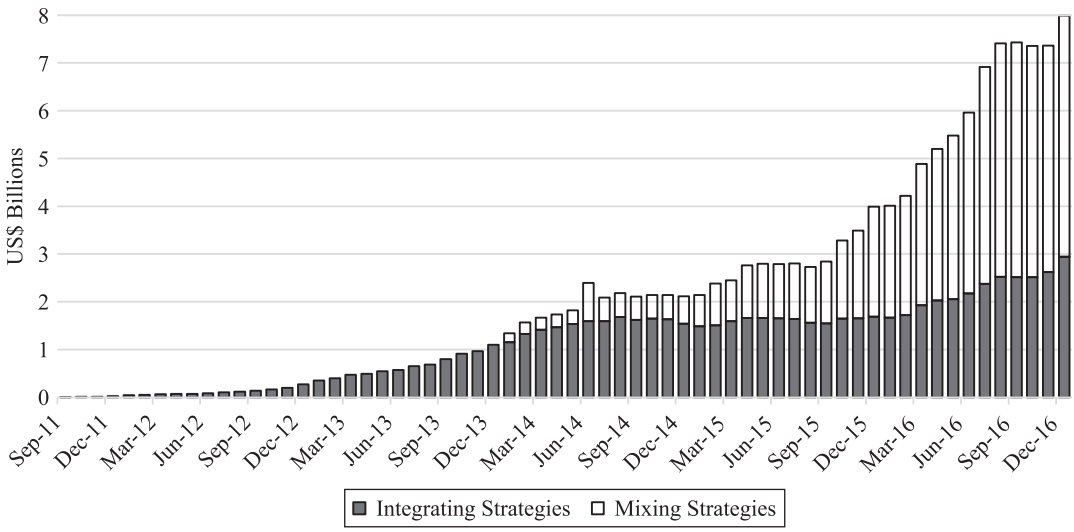
EXHIBIT 1

Total Assets in Multifactor Smart Beta ETFs and Mutual Funds (September 2011–December 2016)



Source: Research Affiliates, LLC, using data from Morningstar Direct.

and mutual fund assets under management (AUM) in multifactor smart beta strategies and the breakdown by portfolio construction process over time. Total assets have grown from near zero in late 2011 to just below $8 billion at the end of 2016. Initially, all of the assets were invested in integrating strategies. In the past three years, however, investments overwhelmingly flowed into the mixing strategies.[2] By the end of the period, the ratio of assets in mixing strategies to assets in integrating strategies was 5:3.

What is causing the discrepancy between theoretical results and actual adoptions? We believe it is driven by preferences that go beyond risk and return. Our research in the United States and developed markets confirms that integrating efficiently creates a multifactor portfolio that delivers exceptional backtested performance. Nonetheless, this method generates portfolio characteristics that smart beta investors who value transparency and low implementation costs do not welcome. First, the integrating approach typically results in a concentrated portfolio with a high turnover rate resulting from rebalancing. These attributes lead to greater active risk, much of which is idiosyncratic, and higher implementation costs. Second, should the investor desire to change the allocation across factors, the integrating methodology would necessitate rebuilding the portfolio from the bottom up. Finally, the integrating approach requires regression analysis to quantify the sources of return. In contrast, the mixing approach facilitates reallocating across factors and is tailor-made for the familiar Brinson model of performance attribution analysis. Thus, investors may favor the mixing approach despite lower expected returns because its simpler methodology and better diversification make for lower trading costs, less unexplained risk, convenient adjustments to factor weights, and easier identification of the contributors to results.

We surmise that the two portfolio construction methods are attractive to different clienteles. Active investors can potentially achieve higher returns with an integrating approach to multifactor investing as part of a sophisticated investment process that includes company-specific risk analysis and skillful trading to minimize implementation costs. Smart beta investors who value intuitive, transparent, low-cost strategies and face stricter internal scrutiny may well prefer the mixing approach.

LITERATURE REVIEW AND OUR CONTRIBUTION

Coincident with the surging popularity of smart beta funds, which Andrew Ang has described as "the vehicle to deliver factor investing" (BlackRock [2018]), a few very recent articles specifically compare mixing

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factor portfolios to constructing long-only multifactor strategies by integrating securities that collectively yield the desired exposures. The authors of these papers variously describe the two approaches as top-down versus bottom-up portfolio construction or combining subportfolios versus building a portfolio of securities. Whatever terminology they adopt, however, they support what we call integrating as the more efficient approach to achieving superior risk-adjusted returns.

Clarke, de Silva, and Thorley [2016] used mean–variance efficiency in assessing portfolio construction methods. As a reference point, they calculated the potential improvement in risk-adjusted return as the difference between the Sharpe ratio of a cap-weighted benchmark and that of a portfolio constructed with unconstrained optimization. Testing four commonly cited factors (low beta, small size, value, and momentum) in the U.S. market, they compared forming long-only multifactor portfolios by integrating securities through optimization against mixing four long-only factor portfolios. Clarke, de Silva, and Thorley found the former captures 90% of potential Sharpe ratio improvement, whereas the latter captures only 40%. Fitzgibbons et al. [2016] reported that integrating is preferable because it favors securities with balanced, positive exposures to multiple factor characteristics and avoids those that have offsetting exposures. On the basis of hypothetical simulations, Fitzgibbons et al. identified three scenarios in which integrating is more effective than mixing: when factors are negatively correlated, when investors have a higher risk tolerance, and when the strategy is open to many factors. Bender and Wang [2016] similarly argued that integrating is superior because of the cross-sectional interaction of factors. They demonstrated through back testing that global portfolios integrating value, low-volatility, quality, and momentum factors outperform their mixing counterparts, especially when pairing uncorrelated factors, such as value and quality or value and momentum.

Basing their arguments on either simulations or backtests, the researchers cited generally overlooked portfolio characteristics that make implementation challenging.[3] Using optimization to construct integrated portfolios tends to cause a high turnover rate: Selecting securities that rank high on multiple uncorrelated factors tends to yield a concentrated portfolio. Concentration and turnover are characteristics associated with high implementation costs (see Chow et al. [2017]). Our principal contribution to the literature is to set forth the practical challenges of the integrating approach. These technical issues have investment consequences: Taking expected implementation costs into account reduces or eliminates the performance advantage of integrating. Given its other benefits (transparency, flexibility in allocation across factors, and ease of performance attribution), mixing may be the more rational approach to multifactor investing.

SUPERIOR HISTORICAL PERFORMANCE OF AN INTEGRATED MULTIFACTOR STRATEGY

We conducted extensive comparisons of integrating versus mixing, with factors constructed in the United States and developed markets using the most widely accepted stock characteristics. Namely, our tests included value, momentum, profitability, investment, and low-beta factors, represented by stocks with the highest book-to-market ratio, the highest trailing one-year return skipping the most recent month, the highest operating profitability, the lowest asset growth, and the lowest market beta, respectively. (see Carhart [1997]; Frazzini and Pedersen [2014]; and Fama and French [2015] for detailed explanations for the factor characteristics.)

To keep our test setup simple, we construct our factors as long-only investable portfolios based on those desirable characteristics. We select the top 20% of the large-cap equity universe by each of the factor characteristics.[4] We then capitalization weight the selected stocks. (We include the results based on an equal-weight scheme in the robustness section.) The value, profitability, and investment factor portfolios are rebalanced annually following Fama and French [2015], whereas the momentum and low-beta factor portfolios are rebalanced quarterly in accordance with widely adopted investment practice.[5]

Our mixing multifactor strategy is an equal-weight combination of those five-factor portfolios. We rebalance the factor allocations to equal weights every quarter. On the other hand, our integrating multifactor strategy is constructed in a way similar to that for individual factor portfolios. We select the top 20% of the large-cap universe by the average factor score.[6] Again, the selected stocks are weighted by market capitalization. The integrating multifactor strategy is also rebalanced quarterly.

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EXHIBIT 2

Simulation of U.S. Single- and Multifactor Strategies (June 1968–December 2016)

	Return	Volatility	Sharpe Ratio	Excess Return	Tracking Error	Information Ratio
Cap-Weighted Benchmark	10.08%	15.5%	0.34	0.00%	0.00%	
Value Factor	11.93%	17.3%	0.41	1.85%	8.12%	0.23
Momentum Factor	11.56%	19.5%	0.34	1.48%	9.91%	0.15
Profitability Factor	10.78%	15.2%	0.39	0.71%	4.58%	0.15
Investment Factor	12.54%	15.5%	0.49	2.46%	5.25%	0.47
Low-Beta Factor	10.65%	11.7%	0.49	0.57%	10.31%	0.06
Integrating Multifactor	13.59%	13.4%	0.65	3.51%	7.89%	0.45
Mixing Multifactor	11.83%	14.3%	0.48	1.75%	3.48%	0.50

Source: Research Affiliates, LLC, using data from Compustat/CRSP.

Although we conducted the tests in both the U.S. and developed markets, we will focus on the U.S. results for the flow of this article. The conclusion highlighted in the following sections equally applies to developed markets. The results for developed markets, based on the comprehensive test in the final section, are presented in the Appendix as a robustness check.

Exhibit 2 shows the long-term return and risk of simulated single- and multifactor strategies. All of the factor strategies outperformed the cap-weighted benchmark, with excess returns ranging from 0.57% to 3.51% per year. As expected, the integrating approach performed exceptionally well: It achieved both the highest absolute return (13.59%) and the highest Sharpe ratio (0.65).

PRACTICAL CHALLENGES IN REALIZING PERFORMANCE

On closer inspection, it appears that the superior performance of the integrating multifactor strategy comes with several characteristics—notably, high concentration and turnover—that compromise its investability.

Exhibit 3 compares the diversification, measured as the number of holdings,[7] of the two multifactor simulations. Over the entire sample period, the integrating strategy is generally less than one-third as diversified as the mixing strategy. The reason for this difference is intuitive. For example, if a security has relatively high return momentum but medium to low exposure to other factors, it belongs to the momentum factor portfolio and therefore enters the mixing strategy at approximately one-fifth of its theoretical maximum weight (the weight it would have if it were held by all factor portfolios).[8] This security thus diversifies the holdings of the mixing strategy, but it is likely excluded from the integrating strategy because of its mediocre average of characteristics.

Exhibit 4 shows that the integrating approach results in consistently higher turnover than the mixing approach. The long-term average turnover rate of the integrating strategy (135.4%) is more than twice that of the mixing strategy (68.0%).[9] Although the concentration of the integrating strategy undoubtedly contributes to the high turnover rate, its portfolio construction process also contributes to this difference. All of the stocks' factor characteristics are reassessed every time the integrating strategy is rebalanced. One of them, the trailing one-year return for momentum, is highly volatile. To maintain meaningful exposure to the momentum factor, the entire integrating strategy is rebalanced quarterly. By contrast, a large proportion of the mixing strategy (related to the value, profitability, and investment factors) can be reconstituted annually. Thus, the mixing strategy affords the practitioner more flexibility to deal with signals that decay with different speeds, whereas a naively constructed integrating strategy must comport with the fastest-decaying signal.

Concentration and turnover are directly related to the cost of implementation. A strategy that holds fewer names and frequently makes concentrated trades incurs higher trading costs. We use the linear framework presented by Aked and Moroz [2015] to quantify

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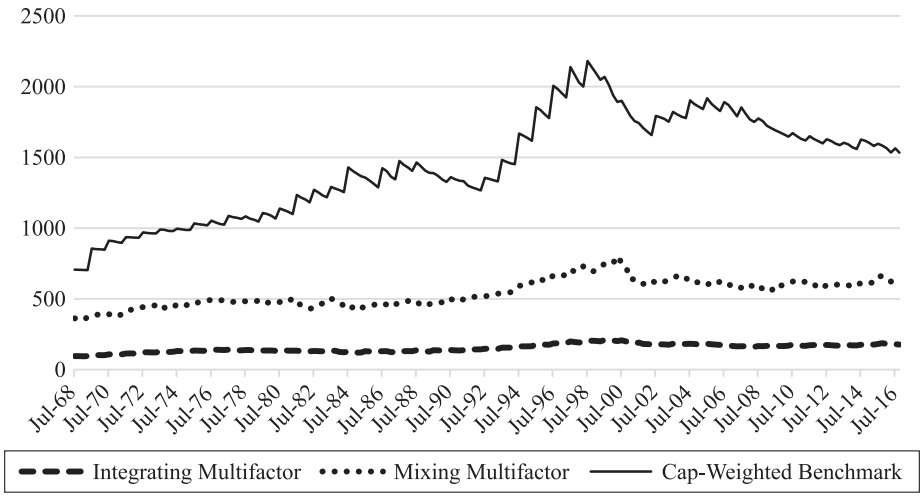
EXHIBIT 3

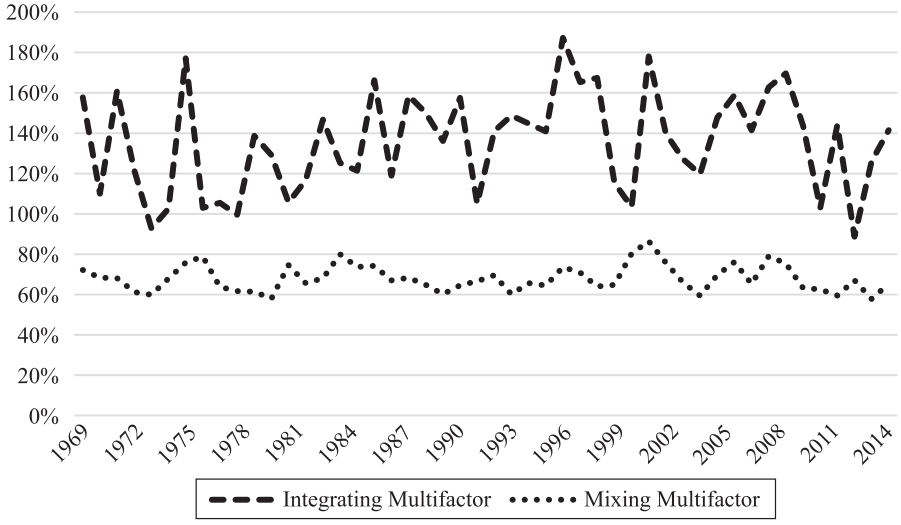
Concentration of U.S. Multifactor Strategies (June 1968–December 2016)



Source: Research Affiliates, LLC, using data from Compustat/CRSP.

EXHIBIT 4

Annual Turnover Rates of U.S. Multifactor Strategies (June 1968–December 2016)



Source: Research Affiliates, LLC, using data from Compustat/CRSP.

the difference in the expected cost of trading: The asset value lost through market impact increases proportionally with the size of the trade. Specifically, the market price of the traded security moves against the trade by 30 bps, on average, for every 10% of the security's daily volume that the trade consumes.

Exhibit 5 reports the expected market impact costs of the factor strategies at an assumed AUM of $1 billion or $5 billion. (This AUM is the aggregate of all assets adopting the strategy.) Because of the high turnover rate and concentration, the momentum factor is the most costly to implement. The integrating multifactor strategy ranks second; at $5 billion AUM, it incurs

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EXHIBIT 5

Expected Costs and Net-of-Cost Performance of Factor Strategies (June 1968–December 2016)

	Cost at $1B AUM (bps)	Cost at $5B AUM (bps)	Gross			Net-of-Cost at $1B AUM			Net-of-Cost at $5B AUM		
			Excess Return	Sharpe Ratio	Information Ratio	Excess Return	Sharpe Ratio	Information Ratio	Excess Return	Sharpe Ratio	Information Ratio
Value Factor	5.3	26.6	1.85%	0.41	0.23	1.80%	0.41	0.22	1.58%	0.39	0.20
Momentum Factor	46.4	231.8	1.48%	0.34	0.15	1.02%	0.32	0.10	–0.84%	0.22	–0.08
Profitability Factor	4.5	22.6	0.71%	0.39	0.15	0.66%	0.39	0.15	0.48%	0.38	0.11
Investment Factor	20.1	100.7	2.46%	0.49	0.47	2.26%	0.48	0.43	1.45%	0.43	0.28
Low-Beta Factor	9.2	45.9	0.57%	0.49	0.06	0.48%	0.48	0.05	0.11%	0.45	0.01
Integrating Multifactor	29.9	149.6	3.51%	0.65	0.45	3.21%	0.63	0.41	2.01%	0.54	0.26
Mixing Multifactor	3.7	18.4	1.75%	0.48	0.50	1.71%	0.48	0.49	1.57%	0.47	0.45

Source: Research Affiliates, LLC, using data from Compustat/CRSP.

estimated trading costs of 149.6 bps per year, more than eight times as expensive as the mixing strategy. Exhibit 5 also shows the performance of the factor strategies net of expected costs. At $5 billion AUM, the difference in rates of return between the integrating and mixing strategies shrinks from 176 bps (3.71% versus 1.75%) to 44 bps (2.01% versus 1.57%), and the difference in Sharpe ratios falls from 0.17 to 0.07.

This cost estimation model assumes all managers tracking the index trade in unison near the end of the rebalance date to minimize tracking error against an index. Active managers who are allowed more latitude can strategically spread out trades to avoid amassing market impact costs. Viewed as a smart-beta indexing strategy, however, much of the return advantage of integrating appears to be washed out at a modest level of adoption.

In addition to the implementation challenges, the high concentration of the integrating strategy results in another unwelcome outcome. As Exhibits 2 and 5 show, the integrating approach, with the highest Sharpe ratio, is more mean–variance efficient than other strategies due in part to its below-market-average return volatility. When evaluating the sources of excess return and active risk, however, it becomes uncertain that integrating is efficient in factor investing. Investors expect consistent excess returns from exposure to robust factors, but we have found evidence that the concentrated integrating approach displays high idiosyncratic risk, casting doubt on its future performance.

We define the percentage of active risk explained by factors as the R^2 from an excess return risk attribution regression:

$$R_t - R^{benchmark_t} \sim \alpha + \beta^M \cdot (Mkt - RF)_t + \beta^s \cdot SMB_t + \beta^H \cdot HML_t + \beta^R \cdot RMW_t + \beta^C \cdot CMA_t + \beta^W \cdot WML_t + \beta^B \cdot BAB_t + \varepsilon_t$$

The dependent variable is the return of the strategy in excess of the return of a cap-weighted benchmark; the explanatory variables represent an augmented factor model, the Fama–French five-factor model plus the momentum and low-beta factors.[10] With the exception of SMB, our strategies explicitly target these factors, and in constructing them, we use the same ranking variables employed in the creation of the factor model. Therefore, little or no model misclassification should be present in our risk attributions.

Exhibit 6 shows the percentage of active risk explained by factors of multifactor strategies at various levels of concentration. They range from very diversified (select the top 75% of stocks) to very concentrated (select only the top 10%). The proportion of active risk explained decreases steadily as the selectiveness of the construction process increases. When the strategy is very concentrated, the percentage of active risk explained drops to 50%; at this point, the unintended idiosyncratic risk is just as prevalent as the systematic factor risk. Exhibit 6 also shows that the target factors consistently explain a lower proportion of the active risk of the integrating strategy than of the mixing strategy.

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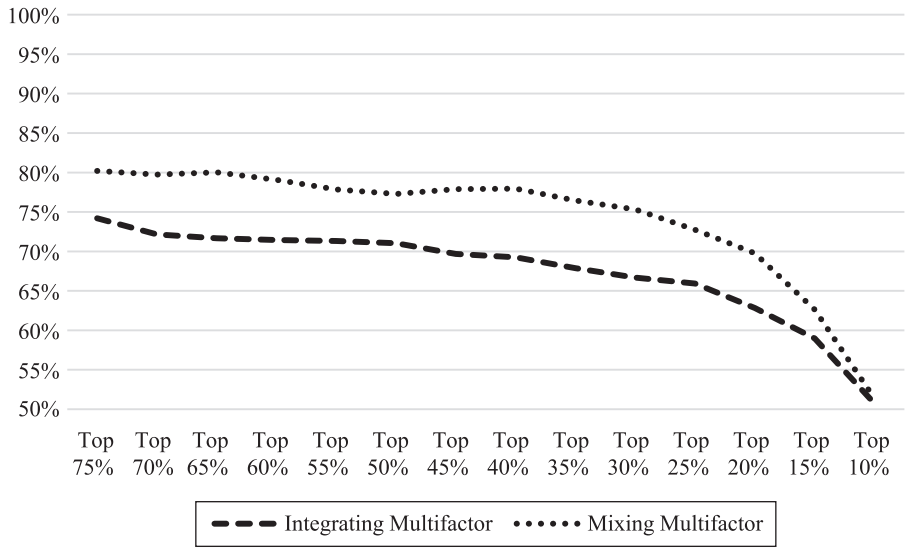
EXHIBIT 6

Percentage of Active Risk Explained by Factors at Various Levels of Concentration (U.S. multifactor strategies, June 1968–December 2016)



Source: Research Affiliates, LLC, using data from Compustat/CRSP.

From this perspective, the integrating approach is less efficient than its Sharpe ratio might indicate because its higher return comes with greater idiosyncratic volatility.

Exhibit 7 shows the decomposition of active risk into factor risk and idiosyncratic risk for the strategies we have examined. In this exhibit, the percentage of risk explained by targeted factors is, as before, the R^2 of the regression based on relevant factors specified in Equation (1). Active risk by targeted factor is the square root of the product of tracking error squared and R^2, and idiosyncratic active risk is the square root of the product of tracking error squared and $(1 - R^2)$. Given that combining individual factors expands the coverage of the mixing portfolio, it is not surprising that the integrating portfolio has a substantially greater tracking error. What stands out is the fact that the integrating strategy's idiosyncratic risk (4.81%) is more than 2.5 times that of the mixing strategy (1.91%). This tells us that a large portion of the 3.51% excess return earned by the integrating strategy comes from sources unrelated to the target factors. We are confident that the premium attributable to robust factors is more likely to persist than the extra performance associated with idiosyncratic risk.

If a strategy is necessarily concentrated, perhaps because it aims for a high expected return, then its company-specific risk should be actively managed with the help of analysts who have expertise and insight at the individual security level and/or a quantitative model that is more sophisticated than a simple factor model. It is less effective to implement such a strategy as a smart beta index.

Although our analysis captures the superior performance of the integrating multifactor strategy in backtested paper portfolios, it also shows the downside of the strategy's higher level of concentration compared to the mixing strategy. Smart beta strategies should be well diversified. They are intended to efficiently provide factor exposures by applying a simple, transparent methodology with low transaction costs to broad cross sections of securities. To assess the suitability of multifactor strategies in a passive and systematic smart beta chassis, we reevaluated their risk and return at higher levels of diversification.

INTEGRATING VERSUS MIXING AT DIFFERENT DIVERSIFICATION LEVELS

Acknowledging that highly concentrated portfolios are expensive to trade and inefficient in providing systematic factor exposures, we relaxed the

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EXHIBIT 7

Decomposition of Active Risk (U.S. strategies, June 1968–December 2016)

	Excess Return	Tracking Error	Factor Attribution								% of Risk Explained by Targeted Factors	Active Risk by Targeted Factor	Idiosyncratic Active Risk
			Alpha	Mkt-RF	SMB	HML	RMW	CMA	WML	BAB			
Value Factor	1.85%	8.12%	–1.85%	0.10		0.71					71.57%	6.87%	4.33%
Momentum Factor	1.48%	9.91%	–2.84%	0.18					0.50		60.29%	7.69%	6.25%
Profitability Factor	0.71%	4.58%	–0.17%	–0.03			0.30				30.06%	2.51%	3.83%
Investment Factor	2.46%	5.25%	–0.67%	0.05				0.62			59.81%	4.06%	3.33%
Low Beta Factor	0.57%	10.31%	–1.05%	–0.41						0.33	57.80%	7.84%	6.70%
Integrating Multifactor	3.51%	7.89%	–1.32%	–0.14	0.06	0.25	0.12	0.15	0.20	0.12	62.83%	6.25%	4.81%
Mixing Multifactor	1.75%	3.48%	–0.84%	–0.04	0.03	0.15	0.03	0.09	0.10	0.04	69.77%	2.91%	1.91%

Source: Research Affiliates, LLC, using data from Compustat/CRSP.

selection constraint. For thorough testing, we varied the selectivity in constructing both the single-factor portfolios and the integrating multifactor portfolio from 5% to 95% with increments of 5% in the total number of stocks. As noted earlier, when the underlying factor portfolios are combined, the mixing multifactor strategy produces a much broader portfolio than the integrating strategy. To ensure comparability, we use the number of holdings in the latest integrating and mixing portfolios as a measure of concentration.

Exhibit 8 compares the return, volatility, Sharpe ratio, value-added, tracking error, and information ratio of the U.S. integrating and mixing multifactor strategies with equivalent numbers of end-of-period holdings.[11] Consistent with our earlier findings, at high levels of concentration, the integrating approach produces higher Sharpe ratios because it has higher return and lower volatility than the mixing approach. For example, with approximately 200 names, the integrating strategy yields a return of 13.58%, volatility of 13.30%, and a Sharpe ratio of 0.65, whereas the mixing strategy has a return of 12.83%, volatility of 15.90%, and a Sharpe ratio of 0.50. In contrast, the considerably higher tracking error of the integrated portfolio (7.20% versus 4.87%) results in a lower information ratio (0.49) than the mixed portfolio (0.57).

With larger numbers of holdings, however, we observe that both paper portfolio returns and transaction costs decrease. We also observe that the empirical advantage of integrating over mixing diminishes. At approximately 600 names, the Sharpe ratio of the mixing and integrating strategies are practically the same (0.49 versus 0.48), whereas the mixing approach registers a higher information ratio (0.50 versus 0.37). With even more holdings, the mixing strategy has better risk-adjusted performance characteristics than the integrating strategy. For instance, at the level of 800 names, the mixing strategy has a return of 11.78%, volatility of 14.20%, and a Sharpe ratio of 0.49, compared to the integrating strategy's return of 10.80%, volatility of 14.70%, and Sharpe ratio of 0.40. We report the same comparison based on the effective number of holdings in the Appendix, where we observe very similar patterns in performance characteristics.

Pursuing this comparison, we took market impact costs into account to reflect real-world implementations. The implementation cost is estimated with the assumption of $5 billion in AUM. Exhibit 9 reports the net-of-cost performance characteristics of the integrating and mixing strategies at similar levels of concentration. As expected, the market impact cost is appreciably greater on less-diversified portfolios. For example, for portfolios with 200 names, the reduction in return due to market impact is about 90 bps for both approaches, whereas it is less than 10 bps for portfolios with 800 names.

The crossing point of Sharpe ratios between the integrating and mixing approaches remains more or less unchanged after considering market impact costs. The integrating multifactor strategy produces a higher net-of-cost Sharpe ratio when the portfolio has fewer than 600 holdings. For instance, at the level of 200 names, on a net-of-cost basis, the integrating strategy yields a return of 12.68% and a Sharpe ratio of 0.59, whereas the mixing strategy has a return of 11.88% and a Sharpe

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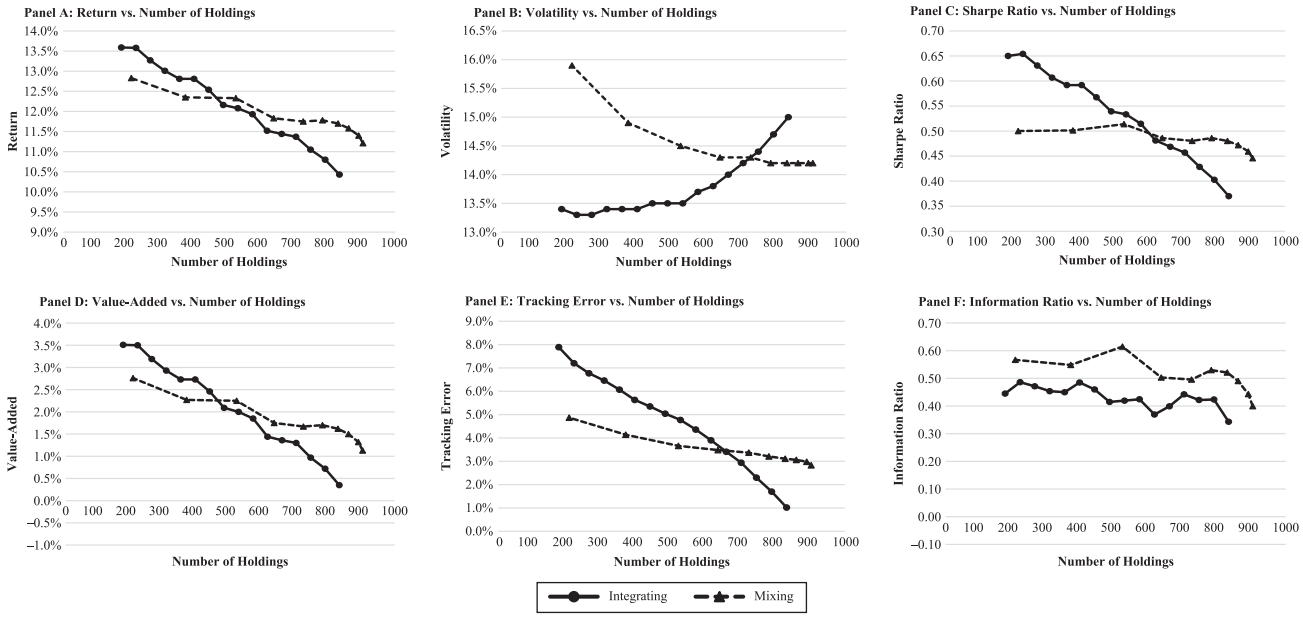
EXHIBIT 8

Comparisons Based on Number of End-of-Period Holdings (U.S. multifactor strategies, June 1968–December 2016)



Source: Research Affiliates, LLC, using data from Compustat/CRSP.

ratio of 0.44. With more holdings, however, the mixing strategy again has both a higher Sharpe ratio and a higher information ratio than the integrating strategy. For example, at the level of 800 names, when net of transaction costs, the mixing strategy has a higher return (11.67% versus 10.77%), higher Sharpe ratio (0.48 versus 0.40), and higher information ratio (0.50 versus 0.41) than the integrating strategy.

As a robustness check, we report, in the Appendix, the results of the same tests using equal-weighted factor portfolios. We generally observe patterns similar to the test results based on capitalization weighting. The crossing point between the Sharpe ratios of the integrating and mixing strategies is around 700 names. For concentrated portfolios, the integrating approach yields better performance characteristics before implementation costs. Because of higher turnover, however, the market impact is much larger than that of capitalization-weighted portfolios. This eliminates the advantage of integrating over mixing. For diversified portfolios, the mixing approach leads to better performance both before and after trading costs. This evidence confirms that our findings are not driven by the choice of a particular weighting scheme.

In the Appendix, we also report out-of-sample robustness tests, which replicate the same multifactor portfolios in developed markets over the period of July 1990 to December 2016. The results are consistent with our findings in the U.S. market. For a highly concentrated portfolio, the integrating strategy has a higher Sharpe ratio than the mixing strategy. For a portfolio containing more than approximately 1,000 names, however, the mixing approach produces both a higher Sharpe ratio and a higher information ratio than the integrating approach.

QUALITATIVE CONSIDERATIONS

The foregoing quantitative analysis of the two approaches to multifactor smart beta investing is informative, but many practical considerations are hard to measure. For example, the governance structure in which practitioners operate may influence investment decisions. Many investment professionals serve as agents for the end investors (principals). It is not enough for an agent and a principal to agree about the factor allocation on the basis of expected returns and risks; the agent must also be equipped to articulate the sources of performance

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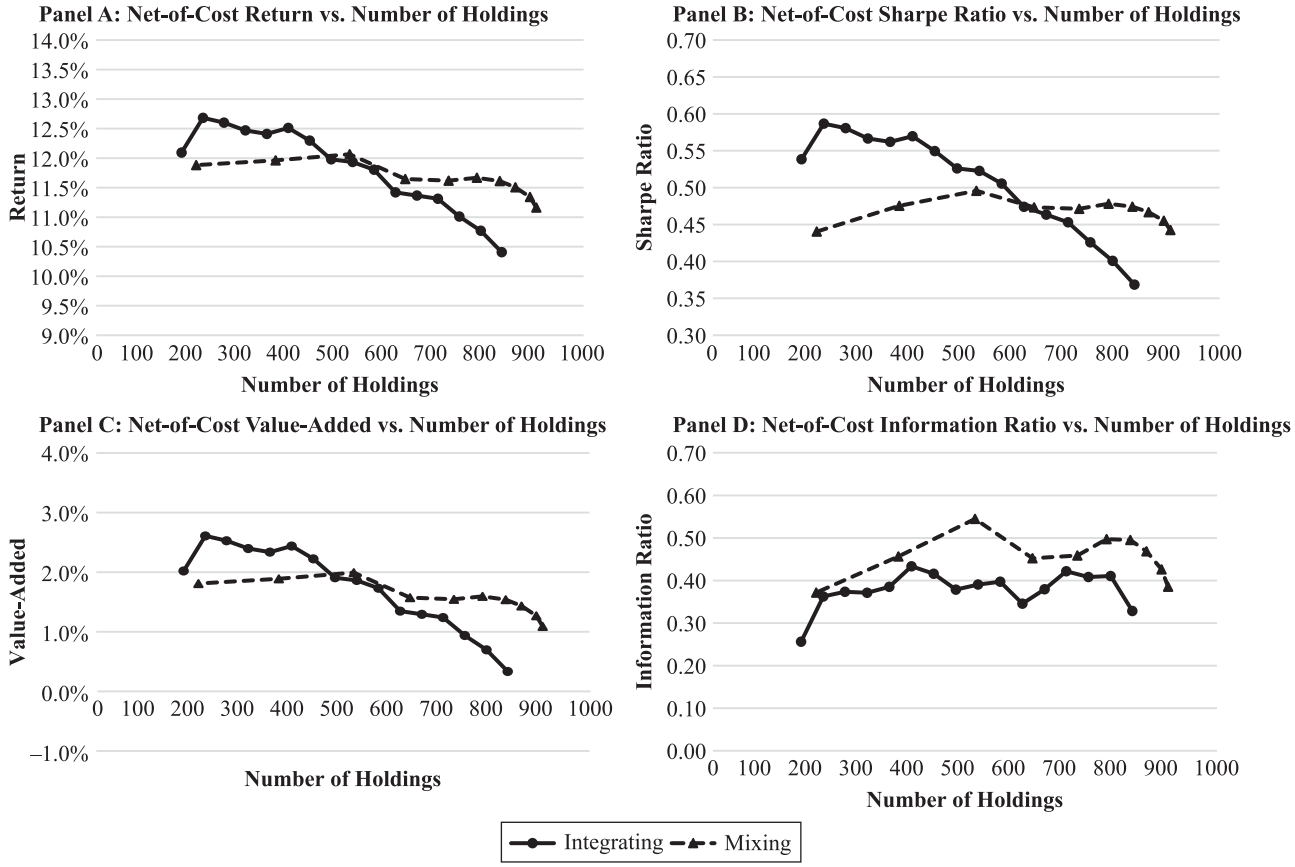
EXHIBIT 9

Net-of-Cost Comparison Based on End-of-Period Number of Holdings (U.S. multifactor strategies, June 1968–December 2016)



Note: Costs are estimated assuming $5 billion in AUM.

Source: Research Affiliates, LLC, using data from Compustat/CRSP.

in terms that make sense to the principal, especially when the portfolio is not doing well. The mixing approach facilitates attribution analysis using a familiar and transparent model. Given its more opaque construction methodology, the relative performance of an integrating portfolio can be more challenging to explain.

A related consideration is the end investor's level of financial and quantitative sophistication. The integrating approach can work well with an investor who clearly recognizes the risks of data snooping and overfitting in backtested results. In addition, as we have already remarked, the integrating approach works better when an investor holds only a subset of the universe, but the resulting concentration leads to higher turnover and, consequently, to higher trading costs, which may wipe out a substantial portion of the performance advantage. Passive execution that focuses on minimizing tracking error against the benchmark tends to bunch trades around the market close on rebalance dates. In the interest of reducing implementation costs, sophisticated principals are more likely to tolerate tracking error that arises from proactive trade management.

CONCLUSION

Integrating and mixing are the two most popular portfolio construction approaches used in multifactor smart beta strategies. The integrating approach (which ranks and selects securities on the basis of a composite score) tends to demonstrate a strong in-sample performance advantage, especially when the selected subset is small and trading costs are disregarded. However, the

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high degree of freedom in computing composite scores makes this methodology vulnerable to overfitting. To help investors form realistic expectations, providers must take conscientious steps to avoid data-snooping bias in their product development practices. In addition, given the relatively higher implementation costs associated with concentrated portfolio holdings, it is important to have sophisticated end investors who can tolerate tracking error and practitioners who can take tactical advantage of the available liquidity in the market to lower the trading costs. The performance advantage associated with the integrating approach then can be preserved.

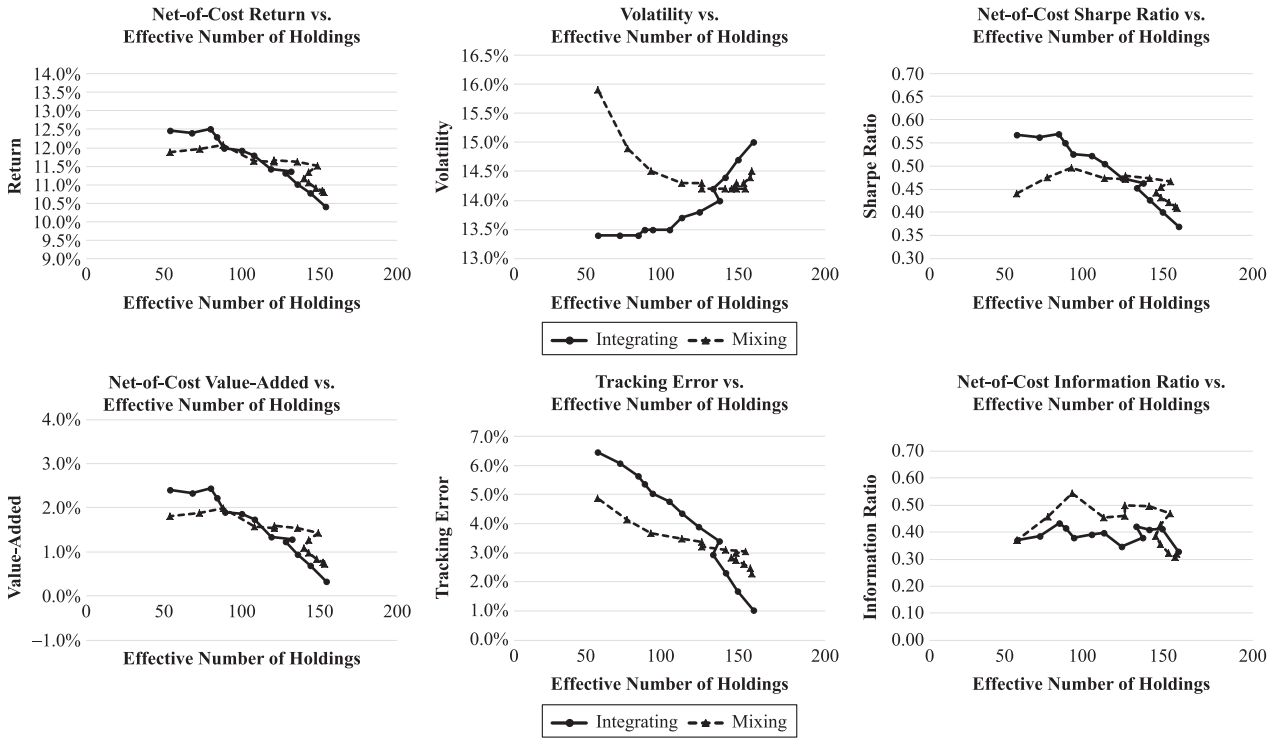
The mixing approach (in which the ranking and stock selection are done with individual factor-specific signals to form stand-alone subportfolios to which a weighting scheme is then applied) exhibits stronger performance when the portfolio is more broadly diversified. Because the trades partially cancel one another out across factor sleeves, the multifactor strategy created this way does not experience as much inefficient trading, and the broad coverage of the portfolio holdings tend to reduce market impact costs further. This more flexible approach offers a greater degree of transparency, and it is easier for investors to understand the performance drivers when different factors deliver different returns. We recommend the mixing approach for multifactor smart beta index construction because of its broadly diversified holdings and easier implementation as well as its transparency and simplicity from a governance perspective. The integrating approach, on the other hand, is more suitable for systematic active strategies.

Appendix

ROBUSTNESS TESTS

Exhibit A1

Net-of-Cost Comparisons Based on Effective Number of Holdings (U.S. multifactor strategies, June 1968–December 2016)



Note: Costs are estimated assuming $5 billion in AUM.

Source: Research Affiliates, LLC, using data from Compustat/CRSP.

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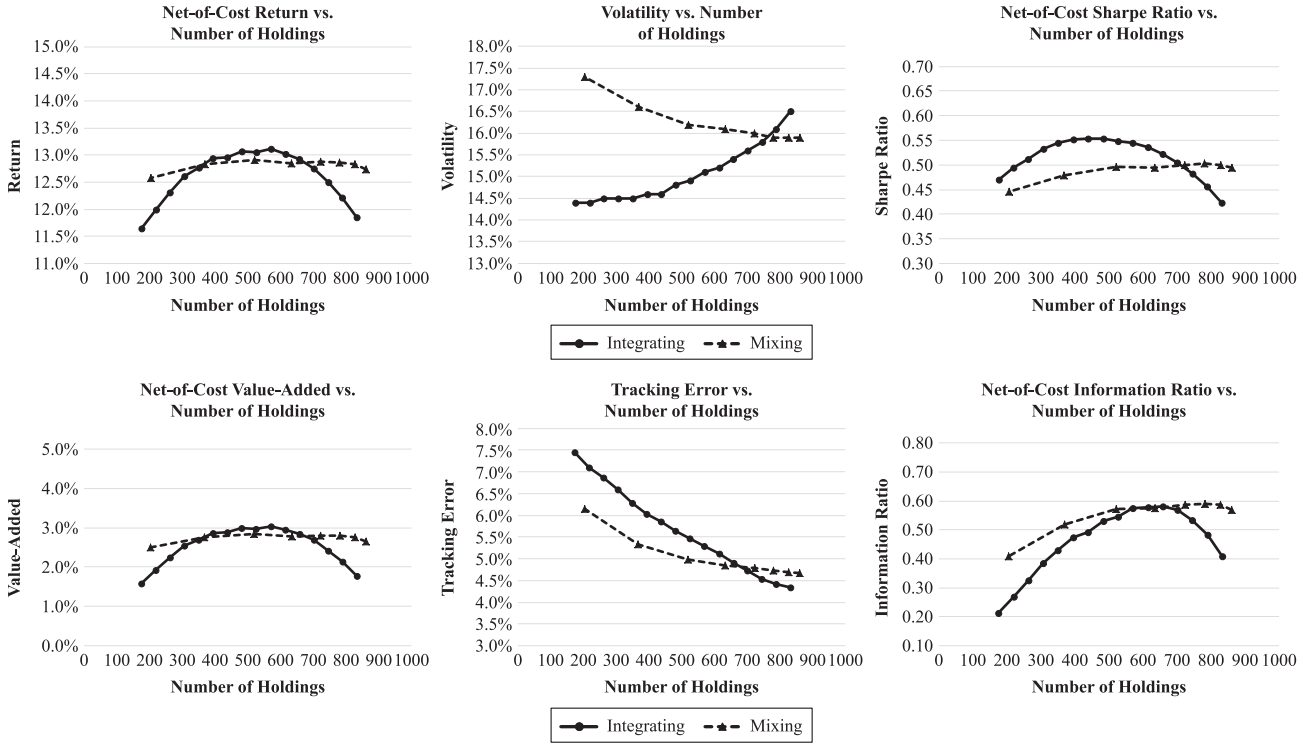
EXHIBIT A2

Net-of-Cost Comparison Based on Number of Holdings with Equal Weighting (U.S. multifactor strategies, June 1968–December 2016)

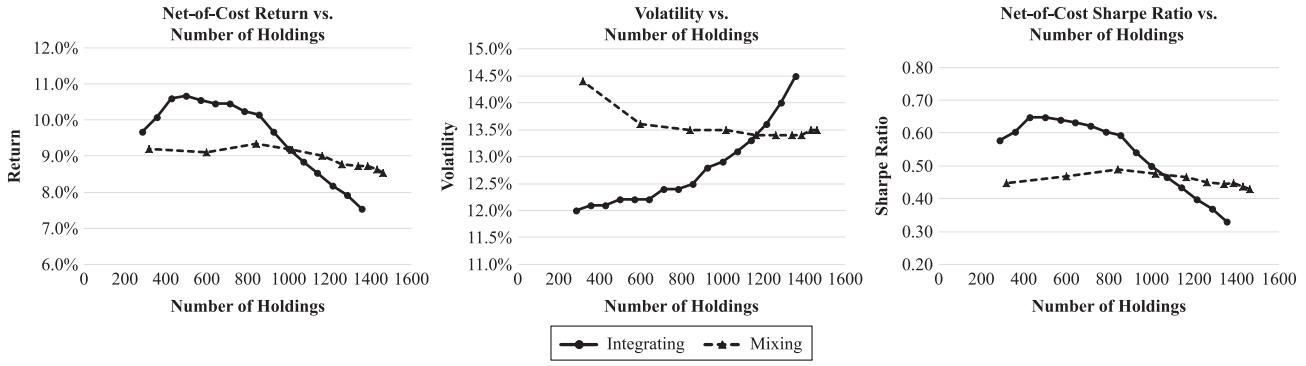


Note: Costs are estimated assuming $5 billion in AUM.

Source: Research Affiliates, LLC, using data from Compustat/CRSP.

EXHIBIT A3

Net-of-Cost Comparison Based on Number of Holdings (developed markets, July 1990–December 2016)



(continued)

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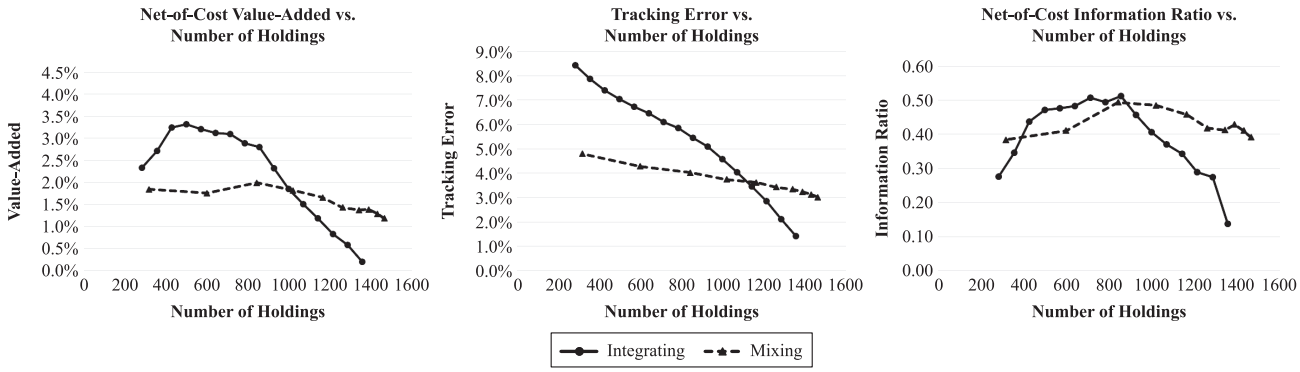
EXHIBIT A3 *(continued)*

Net-of-Cost Comparison Based on Number of Holdings (developed markets, July 1990–December 2016)



Note: Costs are estimated assuming $5 billion in AUM.

Source: Research Affiliates, LLC, using data from Worldscope/Datastream.

ENDNOTES

[1]We used high minus low (HML) and winners minus losers (WML) in the U.S. market from Dr. Kenneth French's Data Library as proxies for the value and momentum factors.

[2]In Exhibit 1, the integrating strategies primarily include the Morningstar Factor Tilt, MSCI Diversified Multiple Factor, and Hartford Multifactor strategies; the principal mixing strategies are the Goldman Sachs ActiveBeta, EDHEC Scientific Beta, MSCI Factor Mix A-Series, John Hancock Multifactor, and J.P. Morgan Equity Risk Premium Multi-Factor strategies.

[3]Fitzgibbons et al. [2016] pointed out that integrating factors improves trading efficiency and reduces turnover by netting out offsetting trades initiated by uncorrelated factors. This is a legitimate implementation-related argument, not for comparing the integrating and mixing approaches but for investing with a single or multiple strategy provider. It is common to have a provider offer mixtures of its own single-factor strategies.

[4]For the U.S. markets, we use the universe of the U.S. stocks from the CRSP/Compustat database to construct our portfolios. Following Fama and French [2015], we define the U.S. large-cap equity universe as stocks whose market capitalizations are greater than the median market capitalization on the NYSE. We then select the top 20% of the large-cap stocks by desired characteristics. For example, to simulate the value factor, we construct the portfolio from the large-cap stocks above the 80th percentile by book-to-market ratio.

For international markets, we use the universe of 23 developed countries from the Worldscope/Datastream database to construct our portfolios. The developed large-cap universe is defined as stocks that represent the top 86% of the entire universe by market capitalization. Factor portfolios are then constructed using the same methodology as for the U.S. factors.

[5]In academic papers, including those by Fama and French [2015] and Frazzini and Pedersen [2014], momentum and low-beta factors are rebalanced monthly. However, because of their high turnover and implementation cost, monthly rebalancing is rarely adopted, and quarterly or even semiannual rebalancing is preferred among investment practitioners. For example, the AQR Momentum Index and S&P 500 Low Volatility Index are rebalanced quarterly, and the MSCI Momentum Index and S&P 500 Minimum Volatility Index are rebalanced on a semiannual basis.

[6]We standardize factor characteristics by calculating cross-sectional Z-scores for each measure. Standardized scores of outliers are capped at ±3. We then take the average of those five-factor scores for each stock.

[7]If diversification is measured by the effective number of holdings, then the integrating strategy is generally less than half as diversified as the mixing strategy. The effective number of holdings is the inverse of the Herfindahl index, which is a measure of concentration computed as the sum of the squared weights of portfolio constituents.

[8]Historically, we find very few stocks that rank well in all metrics.

[9]Our testing did not apply turnover limiting techniques, such as banding, to either strategy.

[10]We obtain the time series of returns for the Fama–French five-factor model (Mkt-RF, small minus big [SMB], HML, robust minus weak [RMW], and conservative minus aggressive [CMA]) and WML from Dr. Kenneth French's Data Library, and the returns of the low-beta factor (BAB) from AQR Capital Management, LLC.

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[11]We report only those strategies that are comparable under both construction methodologies. For instance, the integrating strategy at a 5% selection level is too concentrated, whereas the mixing strategy at a 95% selection level is too diversified. Thus, such extreme strategies are excluded from the figures because they are not appropriate for comparison purposes.

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